

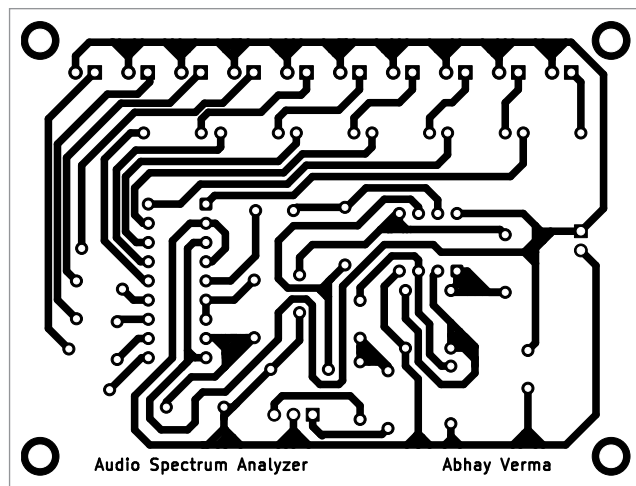


Fig. 3: Actual-size PCB

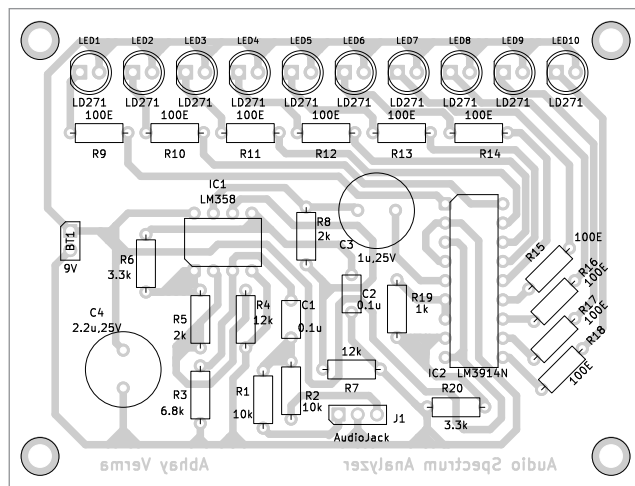


Fig. 4: Components side

PARTS LIST

Semiconductors:

IC1 (IC1A-IC1C) - LM358 dual operational amplifier

IC2 - LM3914 dot/bar display driver

LED1-LED10 - 5mm LEDs

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2 - 10k Ω

R3 - 6.8k Ω

R4, R7 - 12k Ω

R5, R8 - 2k Ω

R6, R20 - 3.3k Ω

R9-R18 - 100 Ω

R19 - 1k Ω

Capacitors:

C1, C2 - 0.1 μ F ceramic

C3 - 1 μ F, 25V electrolytic

C4 - 2.2 μ F electrolytic

Miscellaneous:

J1 - 3.5mm audio jack socket

BATT1 - 9V battery

- Battery clip

- Designed/general-purpose PCB

- IC sockets (8-pin and 18-pin)

- Jumper and connecting wires

can be modified by changing the values of resistors R4, R5, R7, and R8, allowing the device to be optimised for different input signal levels.

At the output of the band-pass filter stage, a 1 μ F capacitor (C3) provides AC coupling. This capacitor blocks any DC component in the signal and ensures that only the amplified AC signal is passed to the next stage, thereby providing a clean, stable input to the display driver.

This filtered and amplified signal is then applied to pin 5 of the LM3914, which senses the input voltage level and drives the LEDs

accordingly. The number of LEDs that glow depends on the amplitude of the filtered signal, which indirectly represents the strength of the selected frequency band in the audio signal.

The LM3914 can operate in two display modes: dot mode and bar mode. In dot mode, pin 9 should be left open. In this mode, only one LED lights up at a time, indicating the signal level. In bar mode, pin 9 should be connected to Vcc. In this mode, multiple LEDs illuminate progressively to form a bar-type display. In this device, bar-mode operation is used because it provides a clearer and more attractive visualisation of the audio spectrum.

Construction and testing

An actual-size, single-sided PCB layout for the audio spectrum analyser is shown in Fig. 3, while the corresponding component layout is shown in Fig. 4. After assembling the circuit on the PCB, mount it inside a suitable enclosure. Mount the 3.5mm audio jack socket (J1) on the front panel of the enclosure and securely fix the assembled PCB inside the enclosure.

Alternatively, the device can be assembled on a suitable general-purpose PCB. First, mount the IC sockets for the LM358 and LM3914 to avoid damage during soldering. Next, solder all the resistors and small components according to the circuit

diagram, followed by the capacitors and preset potentiometer. Ensure correct polarity while connecting the electrolytic capacitors. After this, mount the ten LEDs in a straight line on the front of the PCB so they are clearly visible when glowing.

Now connect the audio input through the 3.5mm audio jack socket to the input section of the circuit. The operational amplifier stages amplify and condition the small audio signal obtained from the audio source. The processed signal is then applied to the input of the LM3914 display driver IC. Carefully check all connections, especially the LED polarity and the supply lines, before inserting the ICs into their sockets.

For testing, connect a 9V battery to power the device and feed an audio signal from a mobile phone, computer, or music player through the audio jack. As the audio signal varies, the LEDs glow sequentially according to the signal level, forming a visual indication of the sound intensity. Adjust the preset potentiometer, if required, so that the LED display responds properly to the input signal level. When the audio volume increases, more LEDs light up, indicating a higher signal amplitude. **EFY**

Abhay Verma is an electronics engineer at EFY and a passionate electronics hobbyist with a keen interest in practical circuit design and DIY devices.